A11 as an algebraic decomposition oracle: a paper bound (run-20261020T0630Z)

Kind: paper bound from committed rows, not a measurement. It prices no pipeline and reports no S. It answers one question before anyone builds the algebraic oracle of `FEASIBILITY.md` (1.5–3 engineer-weeks): could the cheapest presentation measured so far, A11, ever compete per discrete logarithm with the group-law oracle ecbench already runs?

Inputs (all committed)

- **Group-law arm, measured:** `research/ic_frontier_campaign_20261015/phase1/sessions/PKM20/records.jsonl`: `ic.pipeline`, `prime-kummer:t=8`, `rootfind-mitm:m=3`, p = 946177, r = 473453, 32 verified solves, with the matched `rho.negation` arm (32 verified).
- **Algebraic side:** `research/factor_base_designs_20261015/runs/C1-rescore.jsonl`: F4 row-reduction field multiplications per system for A11 and A0 at m = 3, every system replayed with a matched trace. The trace was identical on p_1 and p_2 (Amendment 2), and A11's t = 4 count differs by $3 \cdot 10^{-6}$ between them, so p_1 counts are used at PKM20's prime.
- **Computation:** native, `examples/a11_oracle_paper_bound.rs` (`cargo run --release --example a11_oracle_paper_bound`). Output: `paper_bound.txt`.

The bound, and why it favours A11

Per relation the algebraic oracle costs F/(e·P) GAE and per discrete logarithm (N + 1)·F/(e·P), with:

- F, the mean F4 row-reduction multiplications per system. Symbolic preprocessing, root finding and lifting are not counted.
- e ∈ {1, 12, 50}, field multiplications per group addition. The comparison uses e = 50, the cheapest for A11.
- P = 1 − exp(−8·C(N, 3)/r), $N = 2^{t-1}$ abscissae on the curve, and one relation per success. At t = 8 this gives 0.997; PKM20 measured about 0.96 relations per trial with 123 abscissae.
- Only N + 1 relations per logarithm: no linear algebra and no setup.
- For t > 5 (not measured), F is held at its t = 5 value although it rose with every measured bit. This is an extrapolation, drawn open and dashed.

The group-law arm's unpriced counters (square roots, lookups, row operations, Legendre symbols, target-guard probes) are priced at 0, 0.1 and 1 GAE each (AGENTS.md §5a). Rho's unpriced canonicalisations are not emitted as a count, so rho is a lower bound and context only; the verdict does not rank against it.

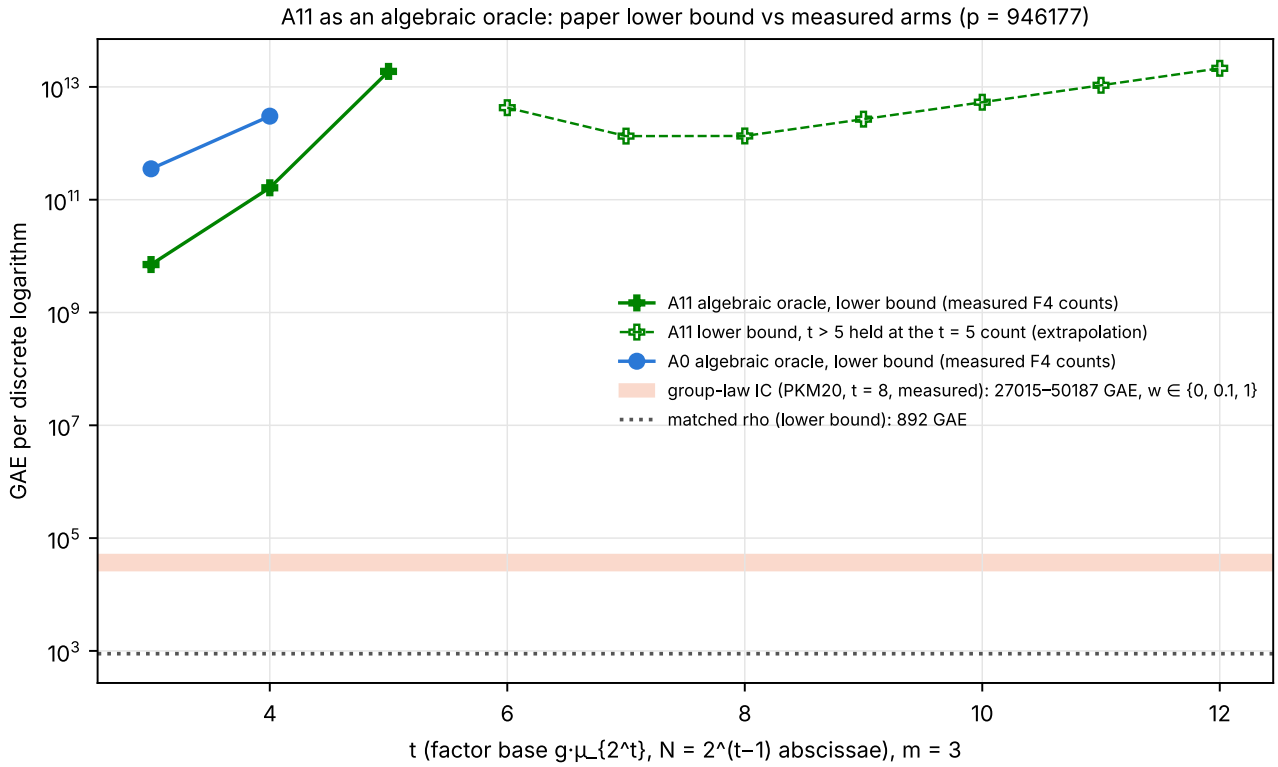


Figure 1: GAE per discrete logarithm against t at m = 3. Filled markers: algebraic lower bounds from measured F4 counts. Open, dashed: A11 with t > 5 held at its t = 5 count (extrapolation). Band: the measured group-law IC arm across w ∈ {0, 0.1, 1}. Dotted: matched rho (lower bound). From `paper_bound.txt`, plotted by `plot.py`.

Numbers (from `paper_bound.txt`)

arm	w = 0	w = 0.1	w = 1

group-law IC, GAE per DLP (mean of 32)	27 015	29 332	50 187
group-law IC, GAE per relation	244.9	265.9	454.6
matched rho, GAE per DLP (lower bound)	892	892	892
cheapest algebraic lower bound (A11, t = 3, e = 50)	$7.085 \cdot 10^{9}$	$7.085 \cdot 10^{9}$	$7.085 \cdot 10^{9}$
ratio algebraic bound / group-law IC	$2.62 \cdot 10^{5}$	$2.42 \cdot 10^{5}$	$1.41 \cdot 10^{5}$

The cheapest algebraic point is A11 at t = 3 (F = $4.79 \cdot 10^{6}$, P = $6.8 \cdot 10^{-5}$). At larger t, P approaches 1 but F grows faster: $8.6 \cdot 10^{8}$ at t = 4 and $5.2 \cdot 10^{11}$ at t = 5. Every A11 and A0 point, measured or held, lies above $7 \cdot 10^{9}$ GAE per logarithm.

Verdict

Across the whole band, even the most generous lower bound for an A11 algebraic oracle costs at least $1.4 \cdot 10^{5}$ times more per discrete logarithm than the group-law oracle ecbench already runs on the same factor base and curve. A misestimate of P or e by three orders of magnitude would not change the order. On this evidence the 1.5–3-week algebraic oracle is not worth building to price A11. The A11 thread closes at “a constant-factor gain inside F4 that does not reach the pipeline”.

Limits

- A bound, not a measurement. Its terms are stated above, each chosen to favour A11.
- One curve size ($p \approx 2^{20}$), m = 3. The group-law arm is at t = 8 only. Its cost per relation is far below any algebraic point, so a t-matched group-law arm would not change the order.
- Rho is a lower bound (unpriced canonicalisations not emitted) and is not ranked against.